Companion LIX: Mass-Mapping Inversion in the Relaxation-Driven Cyclic Cosmology

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Abstract

Mass-mapping techniques reconstruct the projected mass density or gravitational potential from weak-lensing shear fields. In the standard cosmological model, the inversion from shear to convergence and potential is determined by the lensing kernel, the geometry of the Universe, and the statistical properties of the shear field. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the inversion kernel, the reconstruction response, and the coherence of the reconstructed mass maps in a characteristic way. This Companion develops a continuous description of mass- mapping inversion in RDCC, deriving how the relaxation scale influences the reconstruction of κ and Φ, the mode coupling, and the observational signatures in lensing surveys. The resulting framework provides a distinctive probe of the RDCC gravitational field in real-space mass maps.

1 Introduction

Weak gravitational lensing distorts the shapes of background galaxies, allowing the reconstruction of the projected mass density (convergence κ) or the gravitational potential Φ. Mass-mapping techniques invert the shear field to obtain κ or Φ, providing a powerful probe of the matter distribution and the gravitational field.

In the standard cosmological model, the inversion is determined by the lensing kernel, the geometry of the Universe, and the statistical properties of the shear field. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the inversion kernel, the reconstruction response, and the coherence of the reconstructed mass maps in a scale-dependent manner, producing a distinctive signature in mass-mapping analyses.

2 The RDCC Convergence Field

The convergence is related to the gravitational potential by

$$\kappa(\hat{\mathbf{n}}) = \frac{1}{2}\nabla^2\phi(\hat{\mathbf{n}}). \tag{1}$$

In RDCC, the lensing potential evolves as

$$\phi_{\mathrm{RDCC}} = \phi\left[1 - \chi_\phi\left(\frac{\ell}{\ell_{\mathrm{rel}}}\right)^\alpha\right]. \tag{2}$$

The RDCC convergence becomes

$$\kappa_{\mathrm{RDCC}}(\ell) = \kappa(\ell)\left[1 - \chi_\kappa\left(\frac{\ell}{\ell_{\mathrm{rel}}}\right)^\alpha\right]. \tag{3}$$

This modification reflects the scale-dependent evolution of the gravitational potential.

3 Mass-Mapping Inversion in RDCC

The standard inversion from shear to convergence is

$$\hat{\kappa}(\ell) = D(\ell)\,\gamma(\ell), \tag{4}$$

where $D(\ell)$ is the Kaiser–Squires kernel.

In RDCC, the shear field becomes

$$\gamma_{\rm RDCC}(\ell) = \gamma(\ell)\left[1 - \chi_\gamma \left(\frac{\ell}{\ell_{\rm rel}}\right)^{\alpha}\right]. \tag{5}$$

The reconstructed convergence becomes

$$\hat{\kappa}_{\rm RDCC}(\ell) = D(\ell)\,\gamma(\ell)\left[1 - \chi_\gamma \left(\frac{\ell}{\ell_{\rm rel}}\right)^{\alpha}\right]. \tag{6}$$

This modification alters the reconstruction response.

4 Reconstruction Response and RDCC Signatures

The reconstruction response is defined as

$$R_\ell = \frac{\langle \hat{\kappa}_{\rm RDCC}(\ell)\rangle}{\kappa_{\rm RDCC}(\ell)}. \tag{7}$$

In RDCC, this becomes

$$R_{\ell,\rm RDCC} = R_\ell \left[1 - \chi_R \left(\frac{\ell}{\ell_{\rm rel}}\right)^{\alpha}\right]. \tag{8}$$

The relaxation mechanism enhances large-scale coherence and suppresses small-scale structure. Mass maps therefore exhibit:

- enhanced large-scale potential wells,
- suppressed small-scale peaks,
- a characteristic turnover near $\ell_{\rm rel}$,
- modified reconstruction noise at high ℓ.

5 Real-Space Mass Maps in RDCC

The real-space convergence map is

$$\kappa(\hat{\mathbf{n}}) = \int \frac{d^2\ell}{(2\pi)^2}\,\kappa(\ell)\,e^{i\ell\cdot\hat{\mathbf{n}}}. \tag{9}$$

In RDCC, the reconstructed map becomes

$$\kappa_{\rm RDCC}(\hat{\mathbf{n}}) = \kappa(\hat{\mathbf{n}}) - \chi_\kappa \int \frac{d^2\ell}{(2\pi)^2}\,\kappa(\ell)\left(\frac{\ell}{\ell_{\rm rel}}\right)^{\alpha} e^{i\ell\cdot\hat{\mathbf{n}}}. \tag{10}$$

This modification produces a distinctive smoothing pattern tied to the relaxation scale.

6 Conclusion

Mass-mapping inversion in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential and the matter distribution. The relaxation mechanism modifies the inversion kernel, the reconstruction response, and the coherence of the reconstructed mass maps in a scale-dependent manner, producing distinctive signatures in weak-lensing surveys. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of mass reconstruction, nonlinear structure, and the interplay between light and gravity in RDCC.

References

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